CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (Mechanical, AMT)

January 2014

ME – 734 Computer Aided Engineering

Date: 06.01.2014, Monday | **Time: 10:00 a.m. to 01:00 p.m.** | **Maximum Marks: 70**

Instructions:

1. Section I and II must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of scientific calculator is allowed.

SECTION- I

Q1 (a) Differentiate between classical design & computer aided design process. 5

(b) Explain parametric continuity conditions for curves. 3

(c) Explain most common mating conditions for locating and orienting parts in their assembly. 3

Q2 (a) Compare the capabilities of types of displays. 3

(b) Explain the working principle of digitizer. 3

(c) Using Bresenham's line algorithm, find the pixel positions between end points (10, 5) and (15, 9). 6

OR

Q2 (a) What is scan conversion? Why can't a vector scan display produce as many colours as a raster scan display? 3

(b) Differentiate between Analytic curves and Parametric curves. 3

(c) How can you generate a circle with centre 'c' and radius 'r' using Bresenham's algorithm. 6

Q3 (a) Compare geometric transformation with coordinate transformation. Also explain composite transformation. 3

(b) What is homogenous coordinate system? Why is it preferred in computer graphics? 3

(c) A square having end points A(1,1), B(6,1), C(6,6), D(1,6) is rotated by 50° clockwise direction keeping point (6,1) fixed. Find the final coordinates. 6

OR

Q3 (a) Write 3-D transformation matrix for rotation about x, y, z-axis. 3

(b) Prove that 3D rotations are not commutative. 3

(c) Find the reflection of point (5, 5) about a line $y = 3x + 8$. 6

SECTION- II

Q4 (a) What is reverse engineering? Justify its use. 4

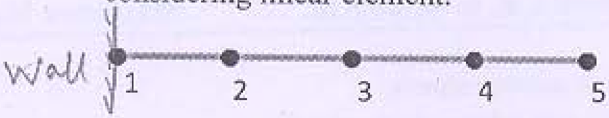
(b) Explain the principle of Rapid Prototyping its advantages and applications. 3

(c) Classify Rapid Prototyping processes and explain any one of them. 3

Q5 In a 1D finite element with two nodes and one temperature degree of freedom at each of the node, derive shape function matrix [N] by assuming suitable temperature function. **7**

OR

Q5 A fin of the length 44 mm is divided in to four equal parts as shown below to analyze the heat transfer. It has five nodes 1, 2, 3, 4 and 5 having temperature 130, 150, 300, 310 and 315°C, respectively. Find the temperature at point 20 mm and 40 mm from the wall by considering linear element. **7**



Q6 Attempt any *three*: **18**

(a) Derive the finite difference formula to compute $d\phi/dx$ using polynomial approach.

(b) A triangular element has nodes at (2, 5), (7, 6) and (6, 10) coordinates having temperature 130°C, 260°C and 390°C respectively. Determine temperature at point (6, 7).

(c) Explain the explicit and implicit approach for temporal discretization with its advantages and limitations

(e) Explain Jacobi method for solving system of simultaneous equation and enlist the steps involved in solution algorithm.

(f) Discuss mathematical and numerical modeling of any one manufacturing process.

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